DS	**Session**	Sorting	**Question Information**	Level 1 Challenge 14

Question description

APPU needed a laptop, so he went to a neighbouring garage sale.

At a sale, there were n laptops.

The cost of a laptop with index i is ai rupees.

Some laptops have a negative price, meaning their owners are willing to pay APPU if he buys their broken laptop. APPU is free to purchase whichever laptop he desires.

Despite his strength, he can only handle m Laptops at a time, and he has no desire to return to the auction.

Please, help APPU find out the maximum sum of money that he can earn.

Constraints:

$1 \leq T \leq 10$

$1 \leq n,m \leq 100$

$-1000 \leq ai \leq 1000$

Input Format:

First line of the input contains T denoting the number of test cases.Each test case has 2 lines :

first line has two spaced integers n m.

second line has n integers [a0...ai...an-1].

Output Format:

The maximum sum of money that LALU can earn, given that he can carry at most m Laptops.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
1
5 3
-6 0 35 -2 4
```

EXPECTED OUTPUT

```
8
```

Test Case 2

INPUT (STDIN)

```
2
6 2
-4 5 0 -1 2 1
7 4
6 1 0 -2 2 -4 3
```

EXPECTED OUTPUT

```
5
6
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int MEGA_SALE(int arr[],int
no,int k)
```

Test Case 2

KEYWORD

```
void bubble_sort(int
arr[],int no)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
320
```

Test Case 3

NLOC

```
55
```

Code

solution.c

```c
#include <stdio.h>

void bubble_sort(int arr[], int no) {
    for (int i = 0; i < no - 1; i++) {
        for (int j = 0; j < no - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int t = arr[j]; arr[j] = arr[j + 1]; arr[j + 1] = t;
            }
        }
    }
}

int MEGA_SALE(int arr[], int no, int k) {
    bubble_sort(arr, no);
    int earned = 0;
    for (int i = 0; i < k && i < no; i++) {
        if (arr[i] < 0) earned += (-arr[i]);
    }
    return earned;
}

int main() {
    int T;
    if (scanf("%d", &T) != 1) return 0;
    while (T--) {
        int n, m;
        scanf("%d %d", &n, &m);
        int arr[n];
        for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
        printf("%d\n", MEGA_SALE(arr, n, m));
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
2
5 3
-6 0 35 -2 4
4 2
-1 -5 2 3
8
6
PS C:\C Programming>
```